

04/09/2026

Ques:- Why chi square is called "sampling distribution"?

Ex:- Let $X \sim N(0,1)$ and define $Y = X^2$. find PDF of Y .

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

CDF of Y : $P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$
 $= P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y})$

$$P(X \leq \sqrt{y}) = \Phi(\sqrt{y})$$
$$= \int_{-\infty}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

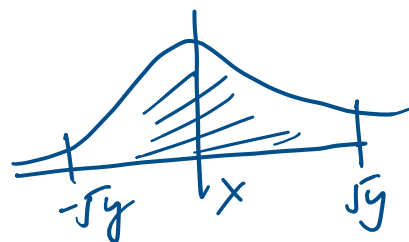
$$P(Y \leq y) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y})$$
$$= \Phi(\sqrt{y}) - 1 + \Phi(\sqrt{y})$$

$$F_Y(y) = 2\Phi(\sqrt{y}) - 1$$

$$f_Y(y) = \begin{cases} \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{\sqrt{y}}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

Now? $\frac{d}{dy} F_Y(y) = 2 \frac{d}{dy} (\Phi(\sqrt{y})) \cdot \frac{d(\sqrt{y})}{dy}$

$$= 2 \underbrace{\frac{1}{\sqrt{2\pi}} e^{-y/2}}_{\text{PDF of } X} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{\sqrt{y}} \quad (1)$$



$$\Phi(-\sqrt{y}) = 1 - \Phi(\sqrt{y})$$

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b(x)) b'(x) - f(x, a(x)) a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f(x, t)}{\partial x} dt$$

$$f(w) = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-w/2} \cdot w^{n/2-1} \rightarrow \text{PDF of chi-square dist}^n$$

Take $n=1$, $f(w) = \frac{1}{2^{1/2} \sqrt{\frac{1}{2}}} e^{-w/2} \cdot \frac{1}{\sqrt{w}} \quad \sqrt{\frac{1}{2}} = \sqrt{\pi}$

$$= \frac{1}{\sqrt{2\pi}} e^{-w/2} \cdot \frac{1}{\sqrt{w}} \quad (2)$$

On comparing (1) and (2), we get $\gamma \sim \chi_1^2$

Thm 2:- Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(0, 1)$. Then $X_i^2 \sim \chi_1^2, i=1, 2, \dots, n$

Then from Thm 1, we get

$$X_1^2 + X_2^2 + \dots + X_n^2 \sim \chi_{1+1+\dots+1}^2 \text{ (n times)}$$

$$\Rightarrow \boxed{\sum_{i=1}^n X_i^2 \sim \chi_n^2}$$

Constants of χ^2 -distⁿ

$$\text{Mean} = \mu_1' = E[X^2] = \int_0^\infty w f(w) dw$$

$$= \int_0^\infty w \cdot \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-w/2} w^{n/2-1} dw$$

$$= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \int_0^\infty w e^{-w/2} w^{n/2-1} dw$$

$$= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \int_0^\infty e^{-w/2} w^{(n/2+1)-1} dw$$

$$\int_0^{\infty} e^{-ax} x^{n-1} dx = \frac{\Gamma n}{a^n} = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \frac{\sqrt{\frac{n}{2} + 1}}{\left(\frac{1}{2}\right)^{\frac{n}{2} + 1}} \quad \Gamma_{n+1} = n\Gamma_n$$

$$= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \frac{\frac{n}{2} \sqrt{\frac{n}{2}}}{\left(\frac{1}{2}\right)^{n/2} \left(\frac{1}{2}\right)} = n$$

Variance:- $\text{Var}(X^2) = E[(X^2)^2] - (E[X^2])^2$

$$= n(n+2) - n^2 = 2n$$

MGF:- $M_{X^2}(t) = E[e^{tX^2}] = \int_0^{\infty} e^{tw} f(w) dw$

$$= \int_0^{\infty} e^{tw} \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-w/2} w^{n/2-1} dw$$

$$= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \int_0^{\infty} e^{-\frac{w}{2}(1-2t)} w^{\frac{n}{2}-1} dw$$

$$= \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} \frac{\sqrt{\frac{n}{2}}}{\left(\frac{1-2t}{2}\right)^{n/2}} = \frac{1}{(1-2t)^{n/2}} = (1-2t)^{-n/2}$$

Suppose we have $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \bar{X} \rightarrow \text{sample mean}$$

$$E[\bar{X}] = E\left[\frac{1}{n} \sum X_i\right] = \frac{1}{n} \sum E[X_i] = \sum_{i=1}^n \frac{\mu}{n} = \frac{n\mu}{n} = \mu$$

$$\text{var}(\bar{X}) = \text{var}\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \sum \text{var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\text{var}(aX+b) = a^2 \text{var}(X) \quad \Rightarrow \quad \boxed{\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$$

$$\# \quad X_1 \sim N(\mu, \sigma^2) \quad X_2 \sim N(\mu, \sigma^2) \quad Y = X_1 + X_2$$

$$M_{X_i}(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$\begin{aligned} M_Y(t) &= E[e^{tY}] = E[e^{t(X_1+X_2)}] = E[e^{tX_1} \cdot e^{tX_2}] \\ &= \exp\left(2\mu t + \sigma^2 t^2\right) = E[e^{tX_1}] \cdot E[e^{tX_2}] \\ &= M_{X_1}(t) \cdot M_{X_2}(t) \end{aligned}$$

$$\# \text{ Define } U_1 = X_1 - \bar{X}, U_2 = X_2 - \bar{X} \dots U_n = X_n - \bar{X}$$

$$\text{and } \bar{U} = (U_1, U_2, \dots, U_n)$$

Thm 3:- The above defined $\bar{X}$ and $\bar{U}$ are statistically independent

Jointly normal + zero covariance $\Rightarrow$ independent

Proof:- $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), U_i = X_i - \bar{X}$

$$\text{cov}(\bar{X}, U_i) = \text{cov}(\bar{X}, X_i - \bar{X}) = \text{cov}(\bar{X}, X_i) - \text{cov}(\bar{X}, \bar{X})$$

$$\text{cov}(\bar{X}, X_i) = \text{cov}\left(\frac{1}{n} \sum_{j=1}^n X_j, X_i\right) = \frac{1}{n} \sum_j \text{cov}(X_j, X_i)$$

$$\because X_i \text{'s are ind.} \Rightarrow \text{cov}(X_i, X_j) = 0 \quad j \neq i$$

$$\Rightarrow \text{cov}(\bar{X}, X_i) = \frac{1}{n} \text{var}(X_i) = \frac{\sigma^2}{n}$$

$$\Rightarrow \text{cov}(\bar{X}, U_i) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0 \quad \Rightarrow \text{cov}(\bar{X}, \bar{U}) = 0$$

Thm 4:- $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \rightarrow \text{sample variance}$

If $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, then $\bar{X}$ and S^2 are independent.
i.e. sample mean and sample variance are independent for a normal random sample.

Proof:- $U_i = X_i - \bar{X} \Rightarrow S^2 = \frac{1}{n-1} \sum U_i^2$

If $\text{Cov}(Y, U) = 0$, then $\text{Cov}(Y, g(U)) = 0$

Here, $Y = \bar{X}$, $g(U) = \frac{1}{n-1} \sum U_i^2$

$\therefore \text{Cov}(\bar{X}, U_i) = 0$, then $\text{Cov}(\bar{X}, \frac{1}{n-1} \sum U_i^2) = 0$
 $\Rightarrow \bar{X}$ and S^2 are independent.

$n=2$

$\bar{X} = \frac{X_1 + X_2}{2}$, $S^2 = (X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 = \frac{(X_1 - X_2)^2}{2}$

$\text{Cov}(X_1 + X_2, X_1 - X_2) = 0 \Rightarrow \text{Cov}(\bar{X}, S^2) = 0$

Thm 5:- If $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, then $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$

Proof:- $X_i \sim N(\mu, \sigma^2)$

$\Rightarrow \frac{X_i - \mu}{\sigma} \sim N(0, 1) \Rightarrow \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2$

Now, $X_i - \mu = (X_i - \bar{X}) + (\bar{X} - \mu)$

$\Rightarrow (X_i - \mu)^2 = (X_i - \bar{X})^2 + (\bar{X} - \mu)^2 + 2(X_i - \bar{X})(\bar{X} - \mu)$

$\Rightarrow \sum (X_i - \mu)^2 = \sum (X_i - \bar{X})^2 + \sum (\bar{X} - \mu)^2 + 2(\bar{X} - \mu) \sum (X_i - \bar{X})$
 $= \sum (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2 + 2(\bar{X} - \mu) \underbrace{\sum (X_i - \bar{X})}_{=0}$

$$\sum (X_i - \bar{X}) = \sum X_i - n \bar{X} = \sum X_i - \sum X_i = 0$$

$$\Rightarrow \underbrace{\sum (X_i - \mu)^2}_{\sigma^2} = \underbrace{\sum (X_i - \bar{X})^2}_{\sigma^2} + \underbrace{n(\bar{X} - \mu)^2}_{\sigma^2}$$

$$\Rightarrow \underbrace{\sum (X_i - \mu)^2}_{\sigma^2} = \underbrace{(n-1)S^2}_{\sigma^2} + \underbrace{n(\bar{X} - \mu)^2}_{\sigma^2}$$

$$X_i \sim N(\mu, \sigma^2) \Rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$$

$$\Rightarrow \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \Rightarrow \frac{n(\bar{X} - \mu)^2}{\sigma^2} \sim \chi_1^2$$

$$\Rightarrow \chi_n^2 = \frac{(n-1)S^2}{\sigma^2} + \chi_1^2$$

We know that if $Y_1 \sim \chi_n^2$, $Y_2 \sim \chi_s^2$ are independent, then $Y_1 + Y_2 \sim \chi_{n+s}^2$

$$\Rightarrow \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Ques:- $E(S^2) = ?$

$$W_1 = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$E\left(\frac{(n-1)S^2}{\sigma^2}\right) = n-1 \Rightarrow \frac{\cancel{(n-1)} E(S^2)}{\sigma^2} = \cancel{n-1}$$

$$\Rightarrow E(S^2) = \sigma^2$$

Ques:-

$$X_i \sim N(50, 16), \quad i=1, 2, \dots, 64$$

$$P(49 < X_0 < 51), \quad P(49 < \bar{X} < 51)$$

Sol:-

$$\begin{aligned} P(49 < X_0 < 51) &= P(49 - 50 < X_0 - 50 < 51 - 50) \\ &= P\left(\frac{49 - 50}{4} < \frac{X_0 - 50}{4} < \frac{51 - 50}{4}\right) \\ &= P\left(-\frac{1}{4} < Z < \frac{1}{4}\right) \\ &= 2P\left(Z < \frac{1}{4}\right) - 1 \\ &= 0.1974 \end{aligned}$$

$$\begin{aligned} P(49 < \bar{X} < 51) &= P\left(\frac{49 - 50}{\sqrt{\frac{16}{64}}} < Z' < \frac{51 - 50}{\sqrt{\frac{16}{64}}}\right) \\ \bar{X} &\sim N\left(50, \frac{16}{64}\right) \\ &= P(-2 < Z' < 2) \\ &= 2\Phi(2) - 1 = 0.9544 \end{aligned}$$